Exam: Classical Test Theory & Reliability

Practice Exam with Solutions | Basic Level

1. [Open question] (3 Pts.) What are the three core assumptions of Classical Test Theory?

2. [Application] (4 Pts.) Scale of 10 items, alpha=0.72. Item 4 alpha-if-deleted=0.78. Remove or keep? What about content validity?

3. [Open question] (3 Pts.) Motor test retest: r=0.95 (1 day), r=0.68 (4 weeks). True reliability?

4. [Open question] (3 Pts.) Split-half r=0.70, 20 items. Spearman-Brown corrected reliability?

5. [Comparison] (3 Pts.) Alpha=0.95 with 40 items vs Alpha=0.80 with 8 items. Which is truly better?

SOLUTIONS

1. X=T+E, E[E]=0, r(T,E)=r(E_i,E_j)=0.

2. Remove improves reliability. Content validity may suffer if item covers unique content. Trade-off.

3. Practice effects inflate. True change deflates. True value ~0.80-0.85.

4. r = 2*0.70/(1+0.70) = 0.824.

5. Not directly comparable. Alpha increases with item count. Quality vs quantity.